

Linear systems and elimination

The square-root upper bound for Gaussian partial-pivoting growth

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging reflects the need for a sharp probabilistic exponent beyond existing polynomial estimates; community impact is its prediction of typical partial-pivoting stability.

Context and notation

All elimination in this problem is in exact arithmetic. Write $\|A\|_{\max} = \max_{ij} |a_{ij}|$. A pivoting path creates successive active Schur complements $S_1 = A, S_2, \dots, S_n$, with row/column permutations as appropriate. Its element-growth factor is

$$\rho(A) = \frac{\max_{1 \leq j \leq n} \|S_j\|_{\max}}{\|A\|_{\max}}.$$

Partial pivoting chooses a largest-magnitude entry in the active first column; complete pivoting chooses one anywhere in the active matrix. If ties occur, a universal statement includes every admissible tie choice; a supremum includes all admissible paths. These conventions remove implementation-dependent ambiguity. Matrices are nonsingular unless otherwise stated.

Problem statement

Let $G_n \in \mathbb{R}^{n \times n}$ have independent $N(0, 1)$ entries. Is the following precise square-root upper-bound conjecture true?

$$\text{For every } \eta > 0, \quad \lim_{n \rightarrow \infty} \Pr\{\rho_{\text{PP}}(G_n) > n^{1/2+\eta}\} = 0.$$

Here growth is measured over the exact-arithmetic Schur complements as defined above. This formulation asks only for the conjectured upper exponent; it does not add an unsupported matching lower-bound assertion or a limiting-distribution claim. It is also distinct from [IE-04](#), which requires a uniform result after perturbing every deterministic center and prescribes an exponential tail.

References

Huang and Tikhomirov, *Average-case analysis of the Gaussian elimination with partial pivoting*, PTRF 189 (2024), 501–567, introduction, discussion of Edelman’s numerical evidence and main theorems. Trefethen and Schreiber, *Average-Case Stability of Gaussian Elimination*, SIAM J. Matrix Anal. Appl. 11 (1990), 335–360.

Earlier status check — 2026-09-08

Searches for *Gaussian elimination* $n^{\{1/2\}}$ 2025 2026 and *site:arxiv.org Gaussian growth factor* 2026 found polynomial upper bounds and the August worst-case results, but no proof at the square-root exponent. The distinction between exact and computed growth factors matters; the catalog statement fixes the former.

Audit update — 2026-09-10

The 2024 [journal article](#) distinguishes its proved polynomial bound from the numerically suggested square-root scale. Searches for Gaussian GEPP growth and later work by Huang–Tikhomirov located no proof of the displayed exponent. A polynomial bound with an unspecified larger exponent does not resolve a parameter range of this sharper target.